

My Contributions (Taken from Regeneron STS Application)

a. Developing / Initiating the purpose of the research (200 word max)	I joined the lab as the preliminary experimentation for the research project began, so Siwei and Dr. Mao came up with the objective of the study, as well as its overall experimental design.
b. Designing the procedures (200 word max)	In calibrating the dissolved oxygen sensor, we had to bring the sample to 100% saturation, then 0% saturation. To rapidly get to 100% saturation, I came up with the idea to use an air pump. To efficiently lower dissolved oxygen levels, I suggested using helium gas instead of nitrogen gas (which came from a different building). Additionally, I communicated that it was important to calibrate the dissolved oxygen sensors before each trial; this was also added to the procedures. To keep the bottles from sliding on the shaker and ensure consistent mixing, I observed that we could tape them down. I came up with the idea to use a syringe for air injections, pushing a specified amount of air into the bottle and taking an equal amount out. Before, dissolved oxygen levels were adjusted by diffusing compressed nitrogen gas into oxygen-saturated nitrified effluent, capping the bottle at the desired DO level. Although this method evolved to use pure oxygen gas rather than ambient air, I played a crucial part in laying out the procedures. I also suggested using carbon footprints to compare the impacts of nitrous oxide and methane emissions—multiplying the masses of greenhouse gases to their global warming potentials.
c. Implementing the procedure (including special techniques or the use of special equipment) (200 word max)	I set up the experiments—collecting woodchips from the Water Research Innovative Facility, inserting woodchips into the bottles, and pouring nitrified effluent into the bottles. To adjust the DO levels, I purged the bottles to 0% saturation using helium gas and injected air for high DO groups. Finally, I cleaned test tubes and other measuring equipment.

d. Gathering / Recording data (200 word max)	To collect data, I gathered samples by operating the syringes, filtering effluent, and conducting dilutions. To gather data, I injected samples into the Gas Chromatograph and helped to set up the Lachat Flow Injection Analysis System. Since data was collected during the school year, I was unable to collect all samples—I went to the lab after school to collect data. Since all data was recorded by Siwei (he was present for all experimentation), he sent me the resulting spreadsheets.
e. Analyzing data (200 word max)	To analyze data, I used the raw data (sent to me by Siwei) to make additional calculations and graphs for my research manuscript. After discussing the best way to depict the data with Siwei, I created the graphs in my manuscript's results section. However, Siwei provided guidance when necessary, teaching me how to do the calculations and giving me the equation for the Gas Chromatograph's calibration curve so I could convert the peak area (automatically measured by the computer program) to $\mu\text{g/L}$.
f. Formulating conclusions (200 word max)	After data analysis, I formulated the conclusions—writing the discussion and conclusion myself. Although I would sometimes ask Siwei specific questions regarding pieces of information I wanted to include or concepts to understand (he provided me with a few textbook chapters to read at this stage so I could better understand the results), the conclusions remained open-ended for me to figure out with the data acquired.